

The Water Cycle

Student sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows **Temperature** (*in degrees Celsius*) and **Humidity** (*relative humidity, in percent*), and below them their shared chart over time. **Pressure does not appear here.** You start with **Start**; you move on with the green button, you go back with **Back**, and the last green button reads **Finish**. Above the chart: **Filter ON/OFF**, **Pause**, **Clear**. **Humidity rises fast (20–40 s) and falls slowly (3–5 min).** **It is not broken.** **The sensor sits 5–10 cm above the water, beside the white steam, never in it:** droplets block it at 100% for the rest of the lesson. Do not blow towards the box.

2. What I do on my own

Water: _____ °C **Weather outside:** _____ **Windows:** ☐ open ☐ closed

First I predict. I do not change my prediction after I measure. **1.** Above the warm water, the humidity will reach _____ %. **2.** It will come down ☐ faster ☐ the same ☐ slower than it went up. **3.** If we warm the air **without adding any water**, the humidity will ☐ rise ☐ fall ☐ stay the same, because _____

The four zones above the vessel, read from the bottom up.

4	COLD SURFACE (the mirror)	Turns to water: condensation	visible? ☐ YES ☐ NO
3	WHITE STEAM above	Droplets of water, already liquid	visible? ☐ YES ☐ NO
2	CLEAR GAP beside the water	Vapour — a gas, like air	visible? ☐ YES ☐ NO
1	LIQUID WATER in the vessel	It all starts here: evaporation	visible? ☐ YES ☐ NO

The zone where **nothing is visible** is zone number ____, and that is exactly where the vapour is.

How much it rises. The times are already written; you may tick **only the arrow**: ↑ rising, ↓ falling, → steady.

Moment	Time	RH (%)	T (°C)	Arrow
Reference, on the bench	0:00			→
Above the warm water	0:30			☐ ↑ ☐ ↓ ☐ →
Above the warm water	1:00			☐ ↑ ☐ ↓ ☐ →
Highest	_____			☐ ↑ ☐ ↓ ☐ →

How much it falls. I moved the sensor away from the water. I start timing again.

Moment	Time	RH (%)	T (°C)	Arrow
After the move	+1 min			☐ ↑ ☐ ↓ ☐ →
After the move	+3 min			☐ ↑ ☐ ↓ ☐ →
After the move	+5 min			☐ ↑ ☐ ↓ ☐ →

The rise. Example: $71 - 44 = 27$ *percentage points* (not percent!).

Mine: _____ - _____ = _____ percentage points.

To the peak: _____ s To return: _____ min How many times slower: _____

The two witnesses. Without them, the measurement above does not prove anything.

Witness	At the start	At the end	Difference (perc. pts)
Above the cold water			
By the radiator, no water			

By the radiator, T ☐ rose ☐ fell, and RH ☐ rose ☐ fell. **Who took the water?**

The dew point (lesson 3): the spoon fogged at _____ °C from the table I get _____ °C

3. The conclusion I write

I predicted that, and the data show that _____

Relative humidity does not say how much water is in the air, but rather _____

What you see above the vessel is not vapour, but rather _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.
